Contrastive DQN in MiniGrid

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created as part of Reinforcement Learning (SoSe 2025)

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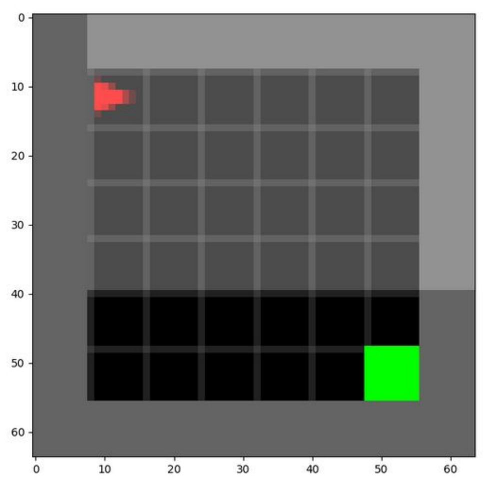
1 TL;DR

- Contrastive encoder for DQN
- Augment Data
- CURL similar
- Tested on Minigrid
  - MiniGrid Empty
  - MiniGrid LavaGap



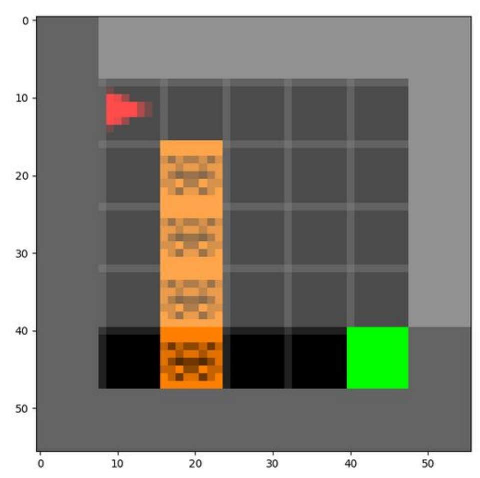
2 Motivation & Problem Setting

Motivation

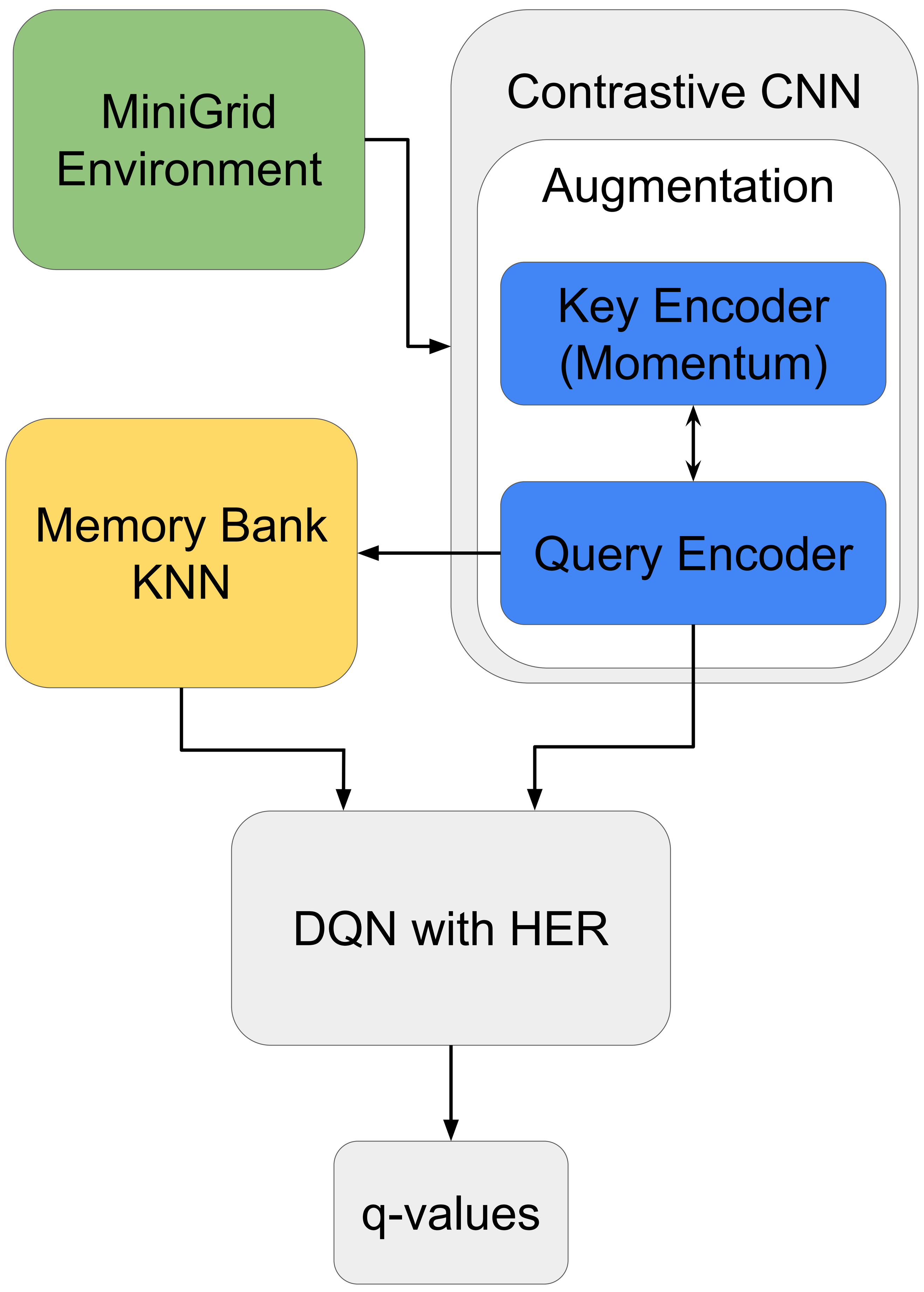
- CURL like contrastive encoder
  [Srinivas et al. 2020]
  - Augment input
  - Query encoder
  - Key encoder (not trained)
- improve information on the statespace

MiniGrid

- Sparse Rewards
- Two Environments
  - easy task
  - hard task

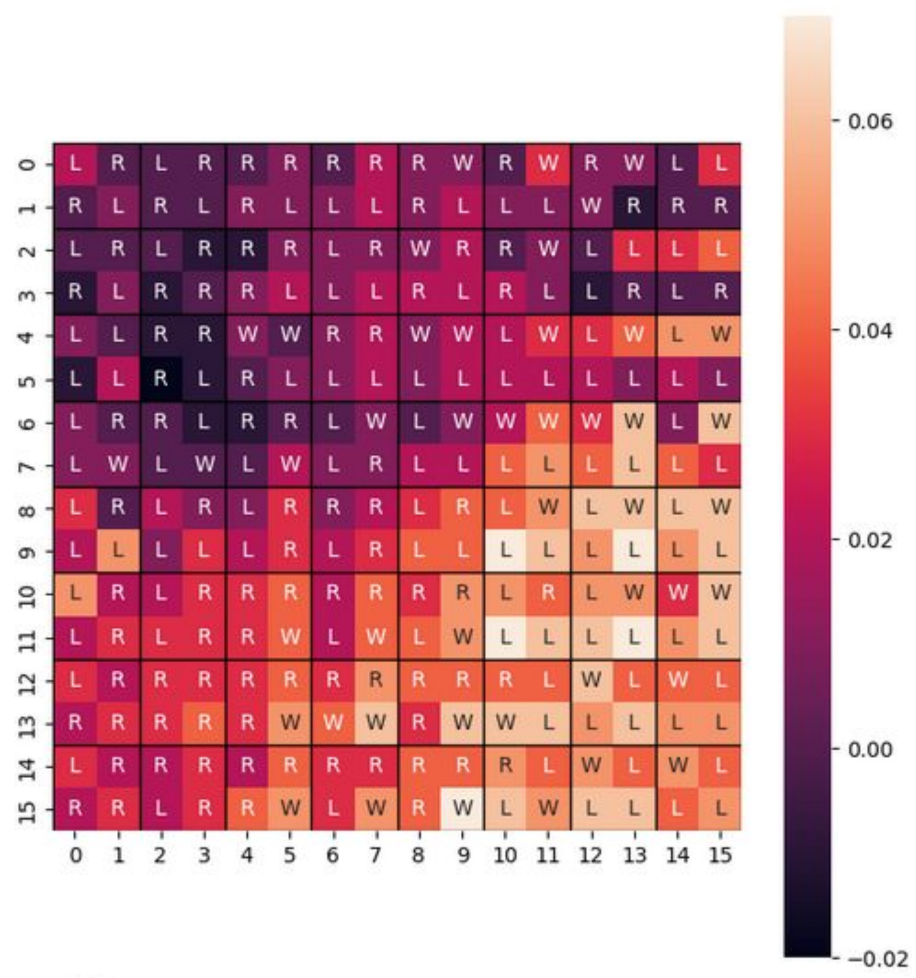


3 Approach

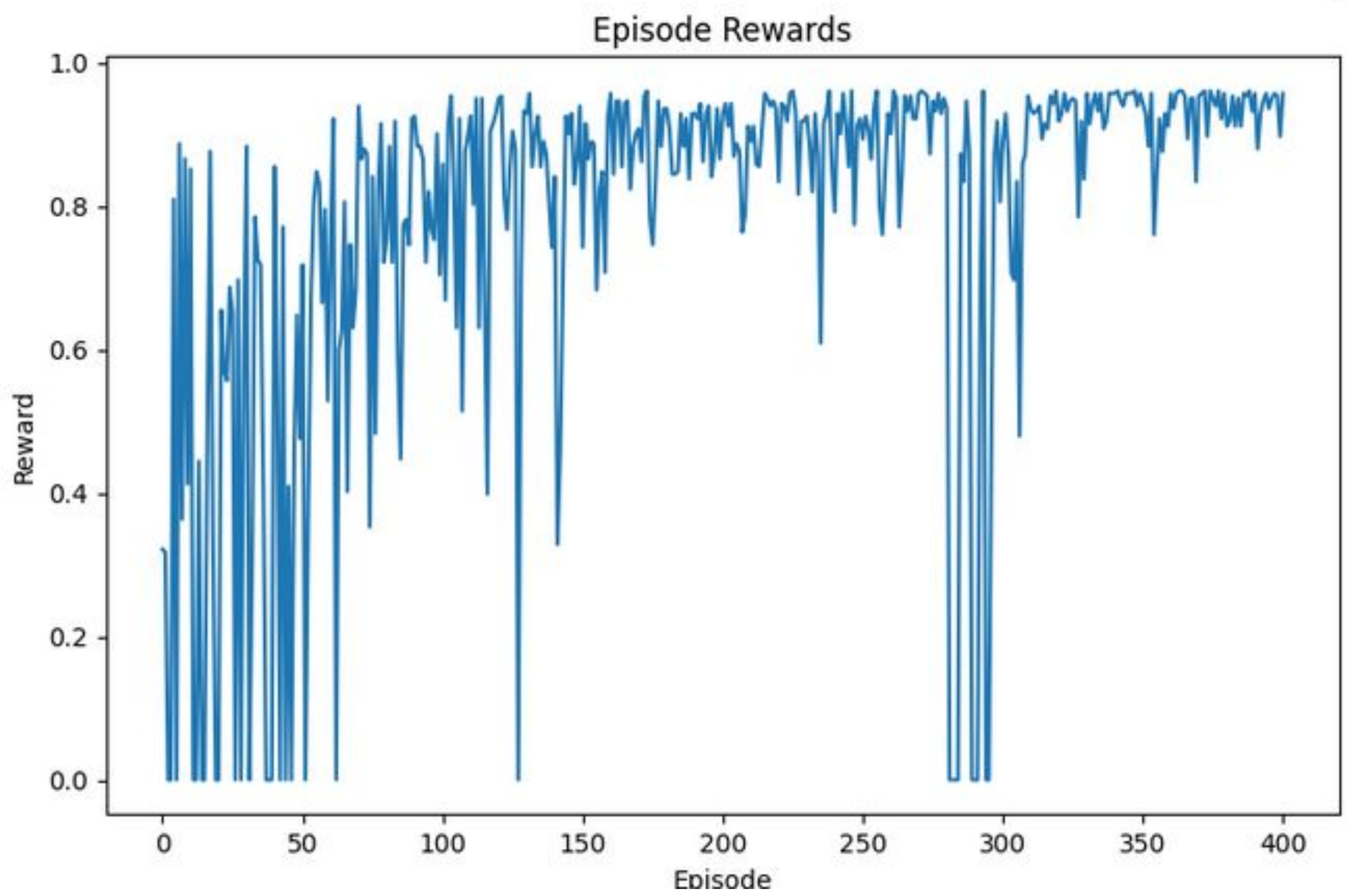


4 Key Insights

HER: set new intermediate goals that give rewards without intrinsic motivation based on transitions



- both RND and our approach had problems on hard tasks
- Our approach learned the easy task well after hyperparameter tuning and the addition of HER



- Our rewards are slightly better than the RND
- Adding HER significantly increases model rewards
  - combating sparse rewards problem

Environment	RND	DQNnoHER	DQNwithHER
MiniGrid-Empty-8x8-v0	0.335 ± 0.021	0.386 ± 0.135	0.842 ± 0.051
MiniGrid-LavaGapS7-v0	0.031 ± 0.006	-	0.141 ± 0.026

Mean Average Reward over 10 seeds per Approach per Level

5 Future Works

- Investigate whether our approach performs differently if the encoder is trained simultaneously with the DQN
- Environments with denser rewards.
- Try more variations of this approach on stronger hardware
- Compare with other approaches using HER